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PARTS LIST

Semiconductors:

IC1 (IC1A-
IC1C) - LM358 op-amp
T1, T2 - BC557 PNP transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R4 - 18k Ω
R2, R5 - 49k Ω
R3, R6 - 680 Ω
R7, R8 - 2k Ω
R9, R11 - 10k Ω
R10, R12 - 100k Ω
RV1 - 10k Ω preset

Capacitors:

C1 - 220 μ F, 25V electrolytic
C2, C4 - 1nF ceramic disc
C3 - 3.7nF ceramic disc
C5, C6 - 10nF ceramic disc
C7 - 2nF ceramic disc
C8 - 10 μ F, 25V electrolytic

Inductors:

L1 - 30 turns of 30AWG wire
wound over 10cm dia.
non-metallic core
L2 - 220 μ H radial inductor

Miscellaneous:

BT1 - 9V battery
BZ1 - Buzzer
- General-purpose PCB/
breadboard
- Connecting/jumper wires
- Enclosure (optional)

(C1) and the virtual-ground bias network (R9 and R11) as close as possible to the LM358 pins to reduce interference. Also, use a common, low-impedance ground bus throughout the circuit to improve frequency stability.

The reference inductor (L2) is a standard 220 μ H radial RF choke, while the search coil (L1) is custom wound using 30 turns of 30AWG enamelled copper wire on a 10-centimetre diameter non-metallic former.

For testing, connect a 9V battery (BT1) and verify the supply voltage. The battery powers both Colpitts oscillators and the LM358 audio amplifier. The mixed RF signal is AC-coupled to the non-inverting input of the LM358 through capacitor C7 (2nF). The operational amplifier is configured as a non-inverting amplifier with an overall gain of about 100. A virtual ground, formed by resistors R9 and R11, ensures symmetrical signal swing for driving the piezo buzzer (BZ1).

After powering the circuit, set the 10k potentiometer (RV1) to its midpoint. Adjust the search coil (L1) by slightly compressing or expanding its windings, or by adding or removing a turn, until a slow, distinct beat frequency is heard. Use RV1 for fine adjustment of the beat frequency and detector sensitivity.

If no beat signal is obtained, the oscillator frequencies are not sufficiently close or may be injection-locked. In that case, slightly reduce the reference oscillator frequency by connecting a small padding capacitor (about 2.2nF) in parallel with its 10nF tank capacitor. Because stray capacitance varies with the PCB or breadboard layout, minor retuning of the coil or tank capacitance may be required. For optimum performance, verify both oscillator frequencies with an oscilloscope and adjust them until they are nearly identical, producing a stable audible beat.

Learning points

This system demonstrates how the beat frequency oscillator (BFO) principle can be used to detect metallic objects using two independent RF oscillators. It explains the importance of frequency matching between the reference and search oscillators and illustrates single-supply analogue signal processing using the LM358 operational amplifier with a virtual ground and high-gain AC amplification. The mixed beat signal is amplified to drive a piezo buzzer, highlighting practical concepts such as signal conditioning, frequency stability, parasitic capacitance, injection locking, and proper power-supply decoupling. Overall, the system provides a solid introduction to analogue circuit design, RF signal mixing, and multi-stage hardware integration in a simple yet effective metal detector. **EFY**

Abhay Verma is an electronics engineer at EFY and a passionate electronics hobbyist with a keen interest in practical circuit design and DIY projects.